

## Practice problem for ANN

Show the mathematical working of Artificial Neural Network by taking the case in figure below. First two columns are the input values for X1 and X2 and the third column is the desired output.

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

Learning rate = 0.2

Threshold = 0.5

Actual output =  $W_1X_1 + W_2X_2$

Next weight adjustment =  $W_n + \Delta W_n$

Change in weight =  $\Delta W_n = \text{learning rate} * (\text{desired output} - \text{actual output}) * X_n$

Show two complete iterations for acquiring the desired output?

| x1 | x2 | w1 | w2 | d | y | $\Delta w_1$ | $\Delta w_2$ |
|----|----|----|----|---|---|--------------|--------------|
|----|----|----|----|---|---|--------------|--------------|

1. Suppose we want to implement a Boolean function  $X_1 \wedge \neg X_2 \wedge X_3$  with a single neuron.
  - a. Calculate the appropriate threshold assuming the weights to be  $W_{X_1} = 1$ ,  $W_{X_2} = -2$  &  $W_{X_3} = 1$ . The truth table of the function is given below. (60 marks)
  - b. What would be the bias weight for this problem, if bias (with a constant input of -1) is used instead of threshold? (40 marks)

Truth Table

| $X_1$ | $X_2$ | $X_3$ | $X_1 \wedge \neg X_2 \wedge X_3$ |
|-------|-------|-------|----------------------------------|
| 0     | 0     | 0     | 0                                |
| 0     | 0     | 1     | 0                                |
| 0     | 1     | 0     | 0                                |
| 0     | 1     | 1     | 0                                |
| 1     | 0     | 0     | 0                                |
| 1     | 0     | 1     | 1                                |
| 1     | 1     | 0     | 0                                |
| 1     | 1     | 1     | 0                                |

**Solution**

| $X_1$ | $X_2$ | $X_3$ | $X_1 \wedge \neg X_2 \wedge X_3$<br>(Output) | $X_1 W_{X_1} + X_2 W_{X_2} + X_3 W_{X_3}$ |
|-------|-------|-------|----------------------------------------------|-------------------------------------------|
| 0     | 0     | 0     | 0                                            | 0                                         |
| 0     | 0     | 1     | 0                                            | 1                                         |
| 0     | 1     | 0     | 0                                            | -2                                        |
| 0     | 1     | 1     | 0                                            | -1                                        |
| 1     | 0     | 0     | 0                                            | 1                                         |
| 1     | 0     | 1     | 1                                            | 2                                         |
| 1     | 1     | 0     | 0                                            | -1                                        |
| 1     | 1     | 1     | 0                                            | 0                                         |

The threshold can be any value  $\theta$  such that  $1 < \theta \leq 2$  (e.g. 1.5)

If bias is used instead of threshold, it can be set at the same value as that of threshold.

## Convolution Neural Networks

A) [8 points] Suppose you have a dataset of images and you want to train a classifier using CNN. For each layer, you need to calculate the size of the associated feature maps, and the number of parameters. The notation follows the convention:

- CONVOLUTION(L, M) represents a convolutional layer with M filters, each them of size  $L \times L$ , Padding and stride parameters are always 1 and 2 respectively.
- POOLING(P) indicates a  $P \times P$  pooling layer with stride P and padding 0.
- FULLY-CONNECTED(N) means fully-connected layer with N neurons.

**n = 512**

| Layer              | Dimensions of Activation map | # of parameters |
|--------------------|------------------------------|-----------------|
| INPUT              | $n * n * 3$                  | 0               |
| CONVOLUTION(7, 16) |                              |                 |
| POOLING(2)         |                              |                 |
| CONVOLUTION(5, 64) |                              |                 |
| POOLING(3)         |                              |                 |
| CONVOLUTION(3, 64) |                              |                 |
| POOLING(2)         |                              |                 |
| FULLY-CONNECTED(5) |                              |                 |

B) Assume the following image and the filter, apply convolution and the pooling.

|                              |  |  |  |  |  |       |  |  |  |  |  |      |  |  |  |  |  |
|------------------------------|--|--|--|--|--|-------|--|--|--|--|--|------|--|--|--|--|--|
| 3 channels of the image are: |  |  |  |  |  |       |  |  |  |  |  |      |  |  |  |  |  |
| 1                            |  |  |  |  |  | 1     |  |  |  |  |  | 1    |  |  |  |  |  |
| 3                            |  |  |  |  |  | 3     |  |  |  |  |  | 3    |  |  |  |  |  |
| 2                            |  |  |  |  |  | 2     |  |  |  |  |  | 2    |  |  |  |  |  |
| 1                            |  |  |  |  |  | 1     |  |  |  |  |  | 1    |  |  |  |  |  |
| 3                            |  |  |  |  |  | 3     |  |  |  |  |  | 3    |  |  |  |  |  |
| Red                          |  |  |  |  |  | Green |  |  |  |  |  | Blue |  |  |  |  |  |

Filter is 3\*3 (same filter for all channels)

|   |   |    |
|---|---|----|
| 1 | 0 | -1 |
| 3 | 1 | 3  |
| 1 | 2 | -1 |

- (i) [6 marks] Apply convolution with Padding and stride parameters as 0 and 2 respectively.
- (ii) [4 marks] Apply Average pooling on the feature map obtained in the previous step.  $1 \times 2$  pooling layer with stride 1 and padding 1.

**SOLUTION IS IN FILE CNN4.PDF**